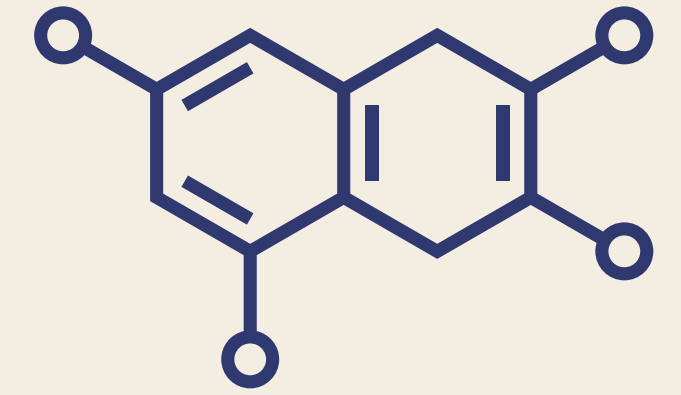


BWSI Medlytics, Group 7



WEEK 1 CHALLENGE: HYPOTHYROIDISM

07/10/2026

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OUR CONTENTS

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PROBLEM

Can machine learning accurately identify hypothyroidism before it is missed?

- Hypothyroidism occurs when the thyroid gland produces too little thyroid hormone
- Symptoms are often vague and overlap with many other conditions
- Laboratory values and patient history are essential for an accurate diagnosis



Why It Matters

- Delayed diagnosis can negatively affect a patient's health and quality of life.
- Early prediction can support faster clinical decision-making.
- Data can be hard to find, and expensive to collect

RESEARCH OBJECTIVES

Which clinical and laboratory variables are most effective for predicting hypothyroidism using machine learning?

Objective 1

Develop Predictive Models

Build and compare multiple machine learning classifiers to distinguish patients with hypothyroidism from those without the disease

Objective 2

Evaluate Model Performance

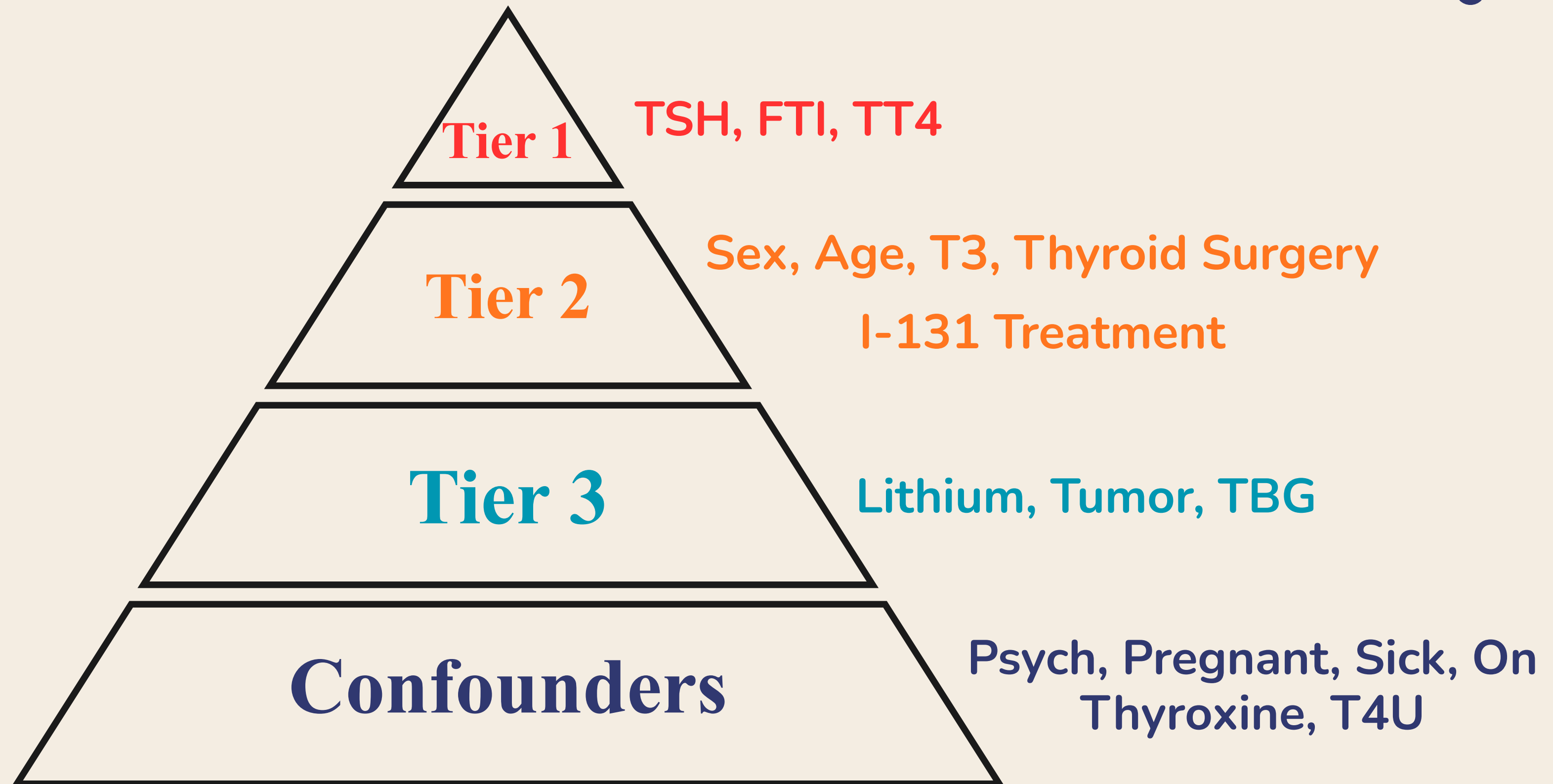
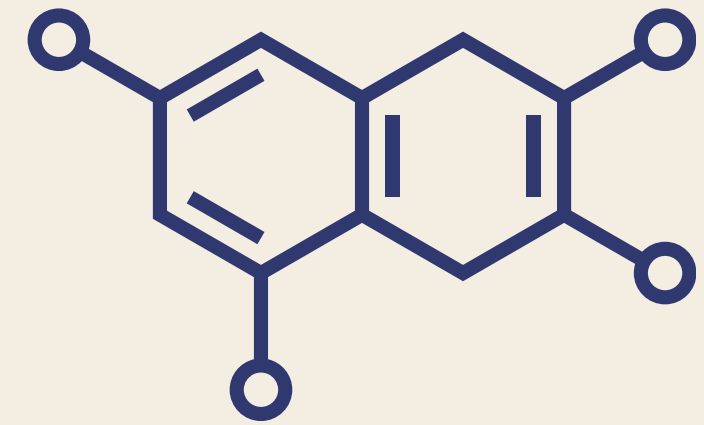
Assess each model using ROC-AUC, accuracy, precision, recall, and confusion matrices to determine predictive performance

Objective 3

Identify Key Predictors

Determine which clinical history and laboratory features contribute most to prediction and compare these findings with established thyroid physiology

FEATURE SELECTION



METHODOLOGY OVERVIEW

Three machine learning models were tested to predict hypothyroidism before deciding on one

Logistic Regression

- Baseline linear classifier
- Highly interpretable
- Estimates probability of hypothyroidism

Random Forest

- Ensemble of decision trees
- Captures complex, non-linear relationships
- Provides feature importance rankings

k-Nearest Neighbors

- Classifies patients based on similar cases
- Simple distance-based approach
- Used as a comparison model

Data Collection

Data Cleaning

- Handle missing values
- Encode categorical features
- Standardize numerical features

Train/Test Split (80/20)

Model Training

- Logistic Regression
- Random Forest
- K-Nearest Neighbors

Model Evaluation

- ROC Curve
- AUC (Validation/Cross-Validation)
- Accuracy
- Confusion Matrix

LOGISTIC REGRESSION MODEL

Model Process:

Hypothyroidism Dataset

Data Preprocessing:

- Clean diagnosis labels
- Handle missing values
- Remove uninformative features
- Convert categorical variables
- Standardize numerical features

Train/Validation Split (80%/20%)

5-Fold Cross Validation (Grid Search)

Logistic Regression (L2 Regularization)

Probability of Hypothyroidism

Grid Search

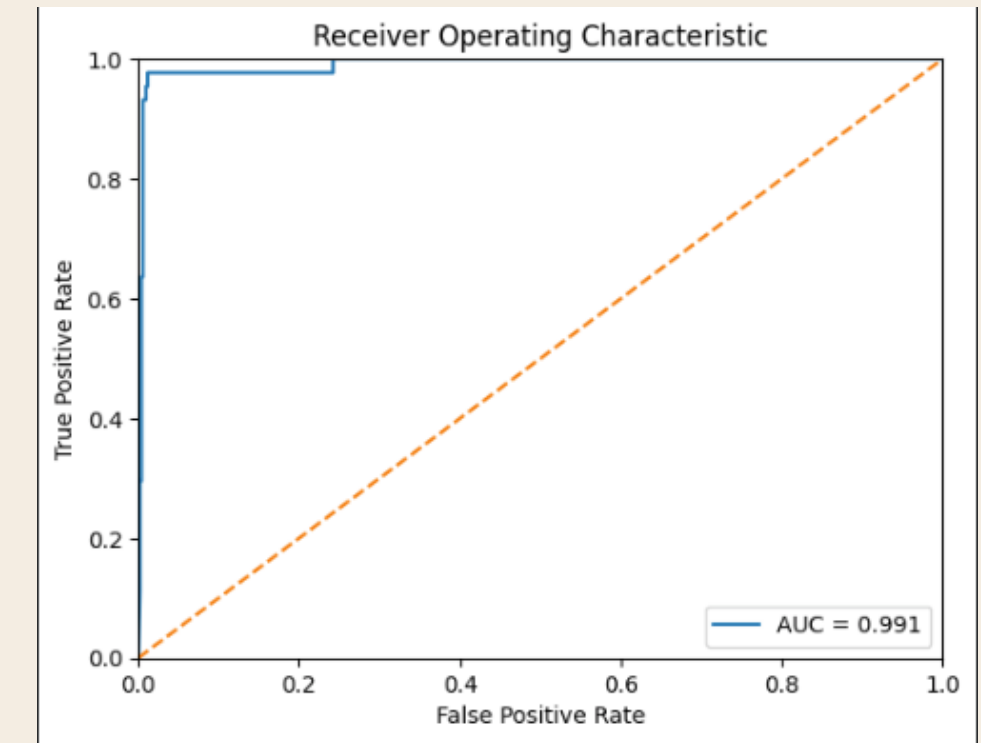
C values tested: 0.01, 0.1, 1, 10, 100

Best C = 10

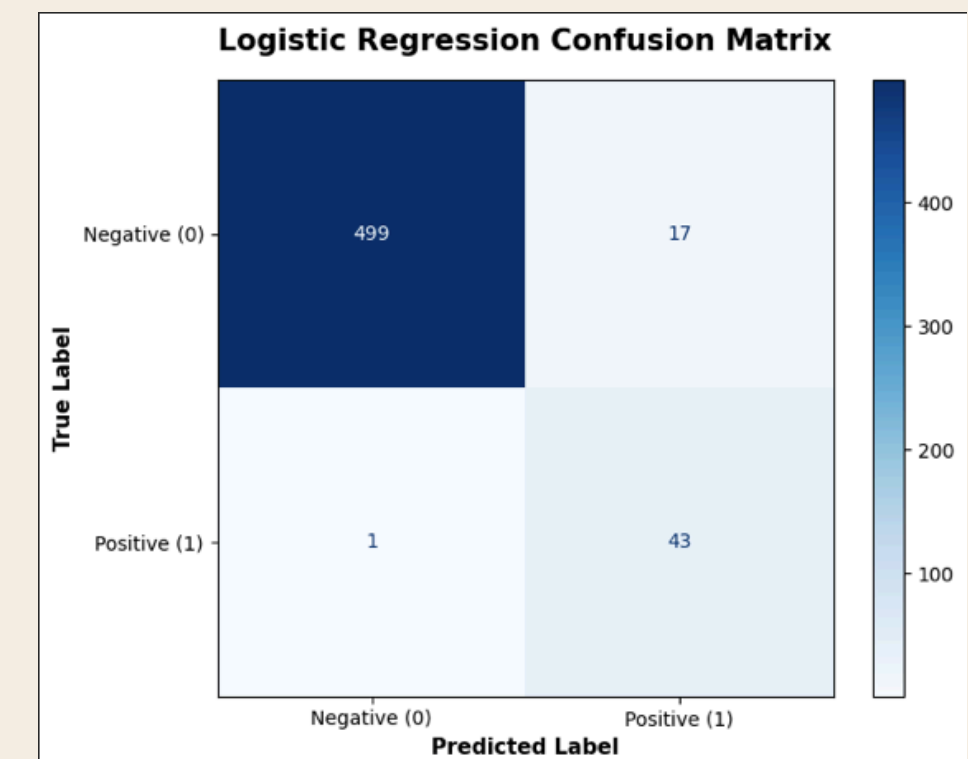
Interpretation & Discussion

- ✓ Thyroid biomarkers (TSH, FTI, and TT4) provided the **strongest predictive signal** for hypothyroidism
- ✓ Logistic Regression combined laboratory, demographic, and clinical features into an interpretable probability of disease
- ✓ L2 regularization and cross-validation improved **generalization** while **reducing overfitting**
- ✓ The model remained highly interpretable, allowing each feature's contribution to be quantified

Results:



Validation AUC: 0.9905
Cross-Validation AUC: 0.980



Random Forest

Model Structure:

1. Separated target label
↓
2. Split train/validation
↓
3. Replaced missing (?) data across training and testing
↓
4. Selected important features across training and validation data
↓
5. Trained a 100-tree Random Forest model with balanced classes

Results:

Confusion Matrix Metric: 1.948

Predicted Healthy	Predicted Hypothyroid
98.3% (True Negatives)	1.7% (False Positives)
0.0% (False Negatives)	100.0% (True Positives)

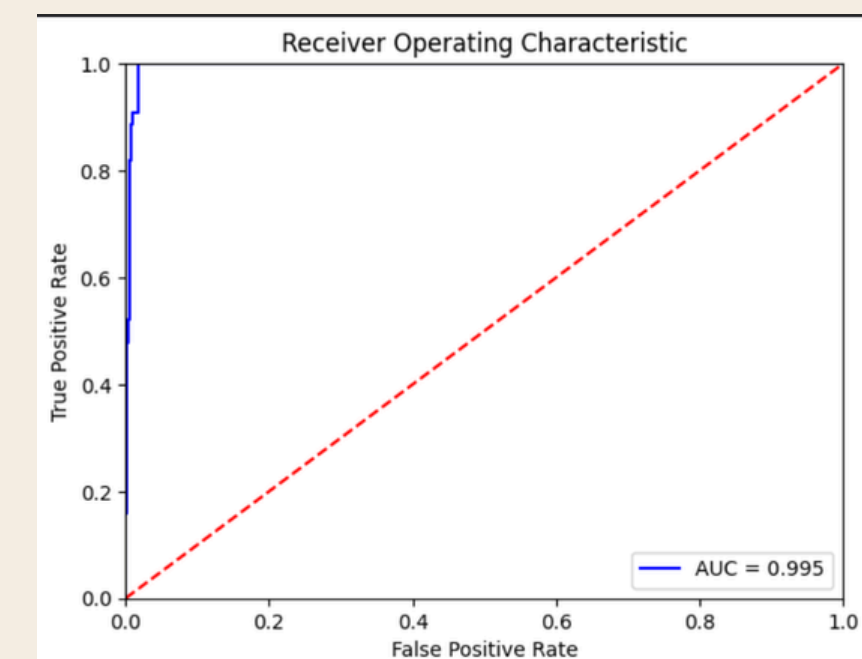
AUC: 0.995

Model:

- `max_depth = 5`
- `random_state = 0`

Interpretation/Evaluation:

- **Decision tree** isolates important features
- **High percentage of healthy:**
Added a balanced weight
- **Controlled tree depth**



NEAREST NEIGHBORS

model process

- clean diagnosis labels
- handle missing values from data
- select main features
+ standardize numerical features
- split data for training/validation
- k nearest neighbors
 - k=9, distance weighted

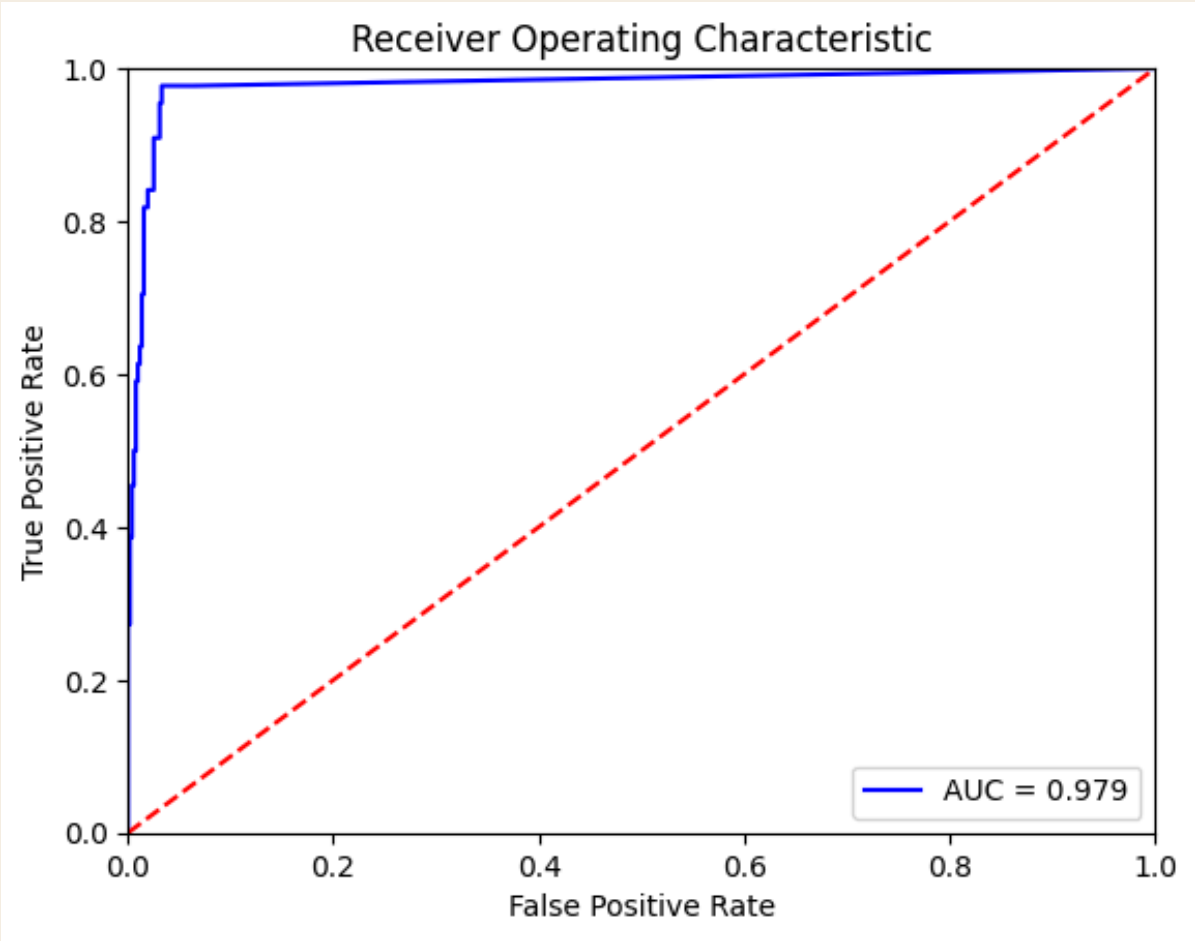
	predicted negative	predicted positive
actual positive	98.4%	1.6%
actual negative	22.7%	77.3%

confusion matrix
metric: +1.27

AUC: 0.979

reducing the number of features improved performance over using full patient records (auc 0.979 vs 0.85)

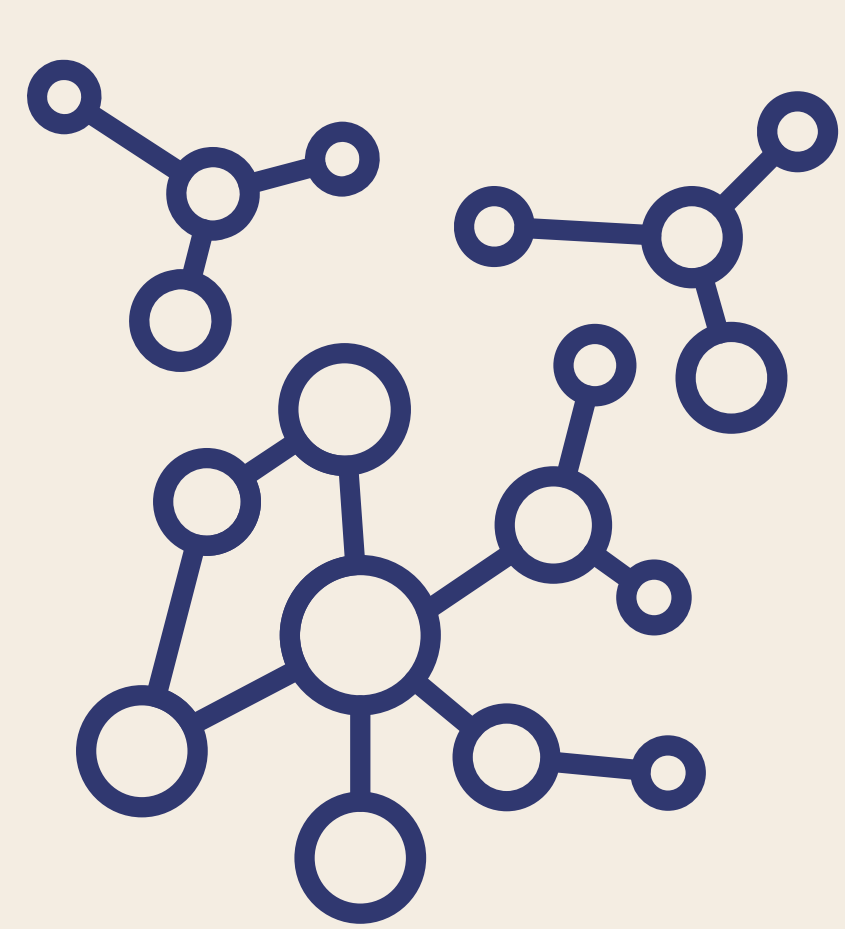
Classifies patients by finding their closest matches among training patients and taking a distance-weighted vote



CONCLUSION/ LIMITATIONS

- **Main goal:** Identify both the best performing classifier and the most useful variables in diagnosis
- **Random Forest Model** had the highest accuracy
- **Limitations:** Handling missing data, risk with unbalanced data





**THANK
YOU!**

